

NOT Gate Design and Simulation Using VHDL

1) VHDL code

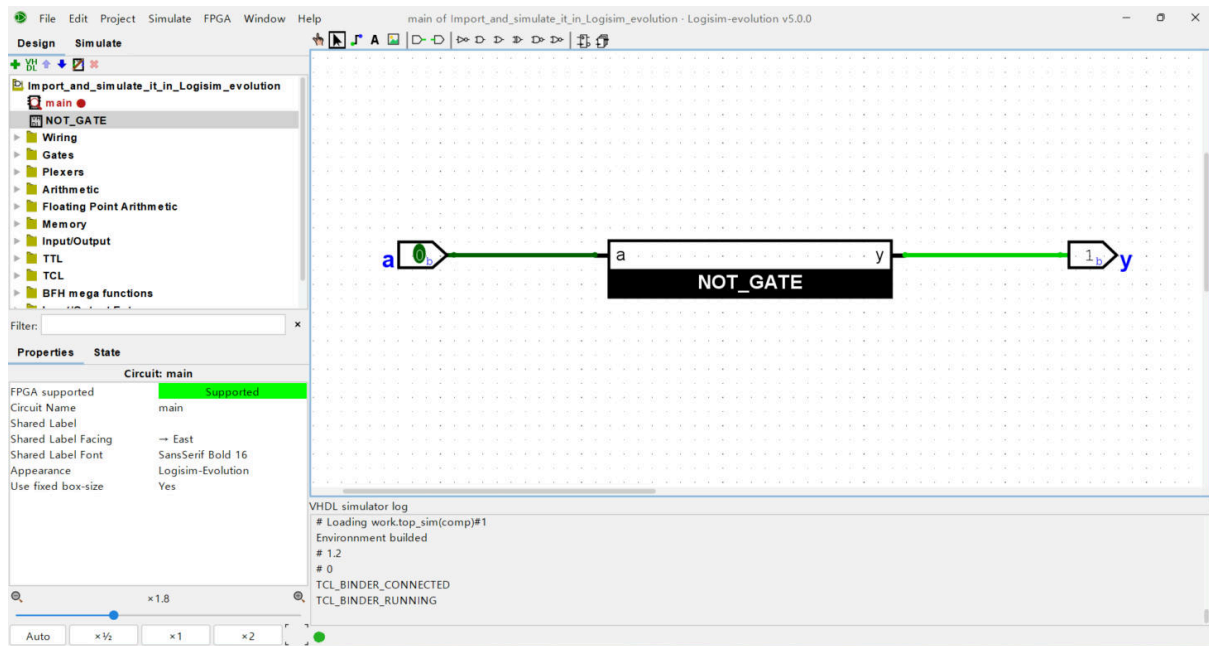
```
LIBRARY ieee;  
USE ieee.std_logic_1164.all;  
  
ENTITY NOT_GATE IS  
  PORT (  
    a : IN std_logic;  
    y : OUT std_logic  
  );  
END NOT_GATE;  
  
ARCHITECTURE TypeArchitecture OF NOT_GATE IS  
BEGIN  
  y <= NOT a;  
END TypeArchitecture;
```

2) circuit

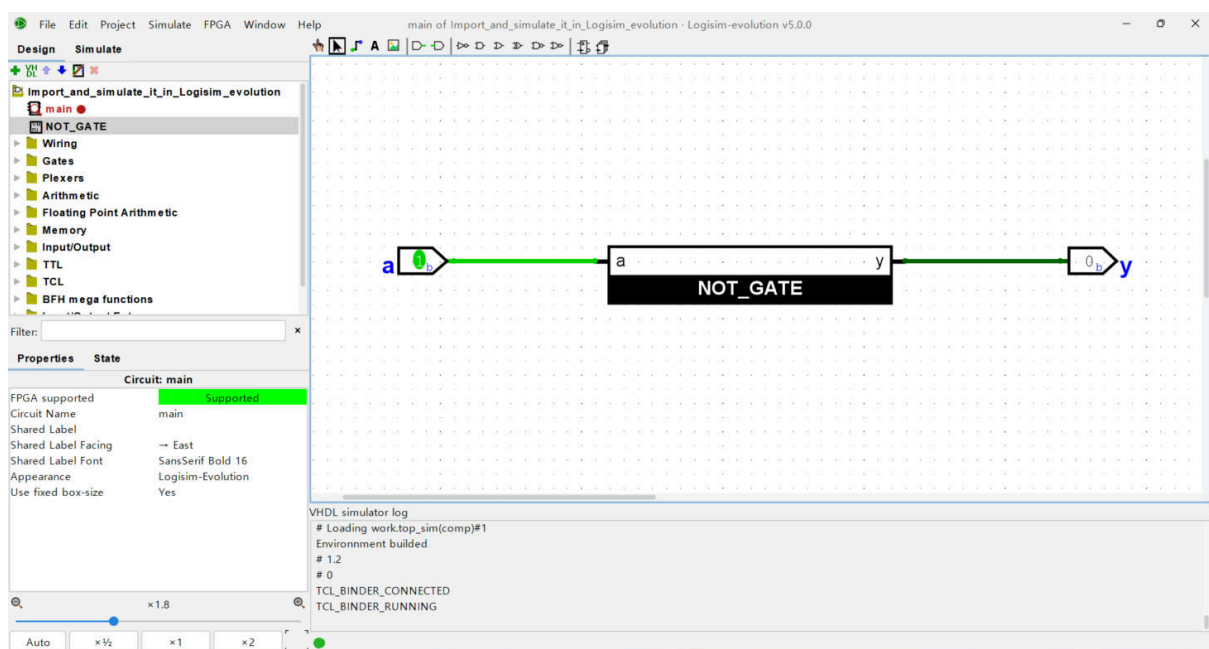


3) simulation screenshots

(A) $a = 0 \rightarrow y = 1$



(B) $a = 1 \rightarrow y = 0$



4) explanation of the results

The NOT gate was simulated in Logisim-evolution with Questa running as the VHDL backend. When input a was set to 0, output y became 1, and when a was set to 1, y became 0. These results match the truth table of a NOT gate, which confirms that the VHDL code works correctly.